

$$J_j \rightarrow \frac{w_j}{l_j}$$

No-ties

We assume that $\frac{w_i}{l_i} \neq \frac{w_j}{l_j} \quad \forall i \neq j$

Output: $\delta = (J_1, J_2, \dots, J_n)$ is the output of our greedy ratio algo.

This implies that $\frac{w_1}{l_1} > \frac{w_2}{l_2} > \frac{w_3}{l_3} > \dots > \frac{w_n}{l_n}$.

If δ is not the optimal sol, then let δ^* be a optimal soln.

Defn. Consecutive Inversion: A pair of jobs (i, j) such that $i > j$ and $\textcircled{i}$ is processed just before $\textcircled{j}$ in some ordering.

$$\hat{\delta} = (\dots \textcircled{i} \textcircled{j})$$

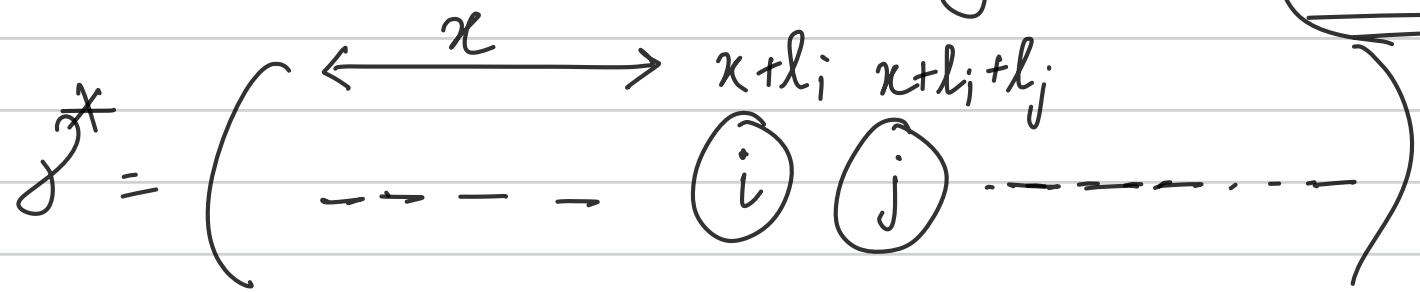
Claim: Every schedule $\hat{\delta}$ other than δ has at least one consecutive inversion. | (Clearly)

$$\delta = (J_1, J_2, J_3, J_4, J_5)$$

$$\hat{\delta} = (J_3, J_5, J_2, J_1, J_4) \quad \underline{(2,1)} \parallel (5,2)$$

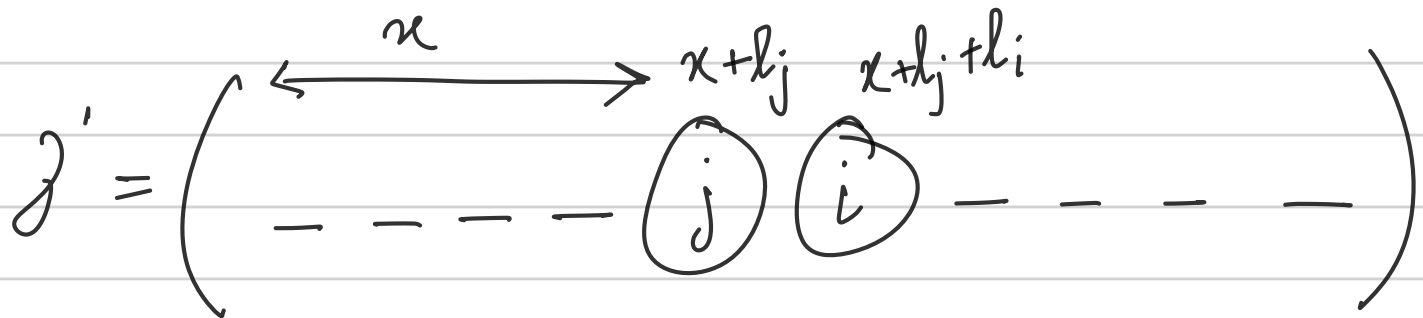
By the above claim, $\mathcal{J}^*$ has atleast one consecutive inversion, say $(i, j) \Rightarrow (i > j)$

$C(\mathcal{J}^*)$



⊗

$C(\mathcal{J}')$



x is the completion time of the job just before i in $\mathcal{J}^*$.

the completion time of the job just before j in $\mathcal{J}'$
 $= x$

$$w_i C_i(\mathcal{J}^*) < C_i(\mathcal{J}') \cdot w_i$$

$\longleftrightarrow$
 $w_i l_j$

$$w_j C_j(\mathcal{J}^*) > C_j(\mathcal{J}') \cdot w_j$$

$\longleftrightarrow$
 $w_j l_i$

⊗ for every other job, wted completion time remains the same.

$$C(\delta') = \sum_{k=1}^n w_k C_k(\delta') = \sum_{k=1}^n w_k C_k(\delta^*) + w_i l_j - w_j l_i$$

$$\Rightarrow C(\delta') = C(\delta^*) + \underline{\underline{(w_i l_j - w_j l_i)}}$$

$$\text{Recall } i > j \Rightarrow \frac{w_j}{l_j} > \frac{w_i}{l_i}$$

$$\Rightarrow \underline{\underline{w_j l_i > w_i l_j}}$$

$$\Rightarrow (w_i l_j - w_j l_i) < 0$$

$$\Rightarrow C(\delta') < C(\delta^*)$$

This contradicts that δ^* is a opt soln.

If $\delta = \delta'$, we can show $\exists$ another δ'' s.t

$$C(\delta'') < C(\delta')$$

$$\underline{\underline{\delta'}} \neq \delta$$

$\Rightarrow \exists$ a consecutive inversion in δ'

$\Rightarrow$ interchanging that consecutive inversion to get δ''

we can prove similarly

In case of tie, say $\frac{w_i}{l_i} = \frac{w_j}{l_j}$

$$\Rightarrow \underline{C(\delta') = C(\delta^*)}$$

(*) δ' is having less consecutive inversion than δ^* .

If $\delta' \neq \delta$, then we can ~~also~~ construct δ'' s.t.
 $C(\delta'') = C(\delta')$

δ'' is having less consecutive inversion than δ' .

$$\delta^* \longrightarrow \delta' \longrightarrow \delta'' \longrightarrow \delta''' \dots \longrightarrow \delta.$$

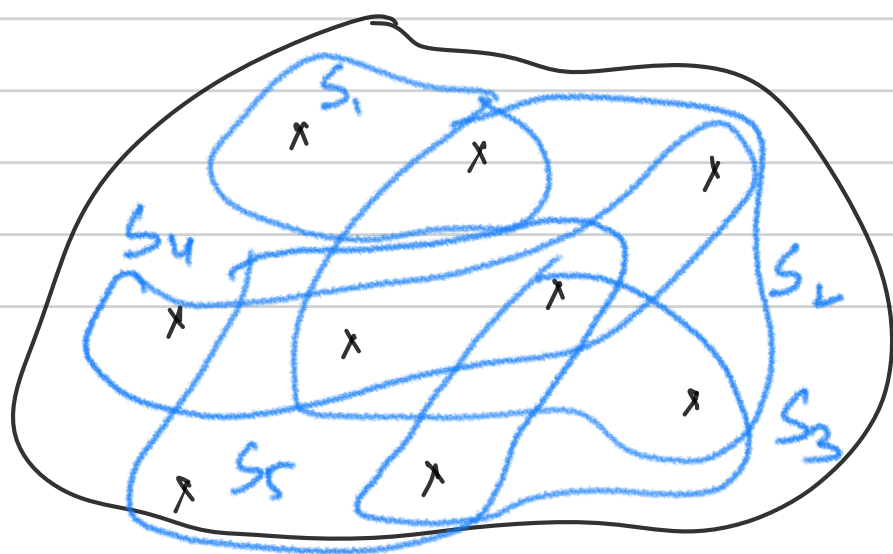
Set-Cover Problem

Input: A set $X = \{x_1, x_2, \dots, x_n\}$ and a family $\mathcal{F}$ of subsets of X , that is, $\mathcal{F} = \{S_1, S_2, \dots, S_m\}$ and $S_i \subseteq X$.

$$\bigcup_{S_i \in \mathcal{F}} S_i = X$$

Output: A subfamily $\mathcal{C} \subseteq \mathcal{F}$ such that $\bigcup_{S_i \in \mathcal{C}} S_i = X$ and has min no of subsets.

←————→
 $\mathcal{C}$ covers X .



$$S_1, S_4, S_5 \\ \cup \\ S_2 \text{ or } S_3$$

Greedy Algo

Natural Choice is to select the set S_t which covers max elements of X .

→ $(X \setminus S_t) \rightarrow$ set of uncovered element.

Choose the set that covers max elements of $(X \setminus S_t)$



Greedy Set Cover Algo

$U = X$ (Uncovered element)

$\mathcal{L} = \emptyset$

while (U is non empty)

{ Select S_i s.t. $(S_i \cap U)$ is max

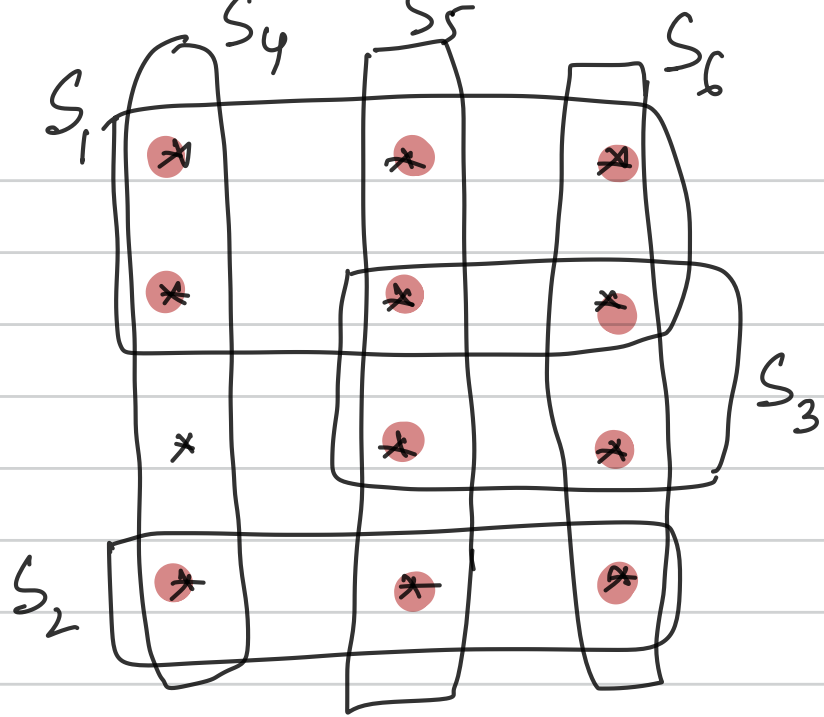
$\mathcal{L} = \mathcal{L} \cup \{S_i\}$

$U = U \setminus S_i$

}

Return $(\mathcal{L})$.

Eg:



$$F = \{S_1, S_2, S_3, S_4, S_5, S_6\}$$

$$\{S_1, S_2, S_3, S_4\}$$

↓
Output of our algo.

$$\text{Opt Soln: } \{S_4, S_5, S_6\}$$

→ Greedy ^{set cover} algo does not give correct soln always.

→ This algo gives a soln which is close to the optimal sol always.

(*) If $\mathcal{C}$ is the output of Greedy set cover algo, and $\mathcal{C}^*$ is a optimal soln of the problem, then

$$\Rightarrow |\mathcal{C}| \leq O(\log n) \times |\mathcal{C}^*|$$

Approximation Ratio.
(Approximation Algo)

$$|\mathcal{C}| \leq 1.000 \times |\mathcal{C}^*|$$